Constant Auxiliary Losses Considered Harmful: A Cautionary Tale in Belief-Learning Multi-Agent Reinforcement Learning

Anonymous Author(s)
Affiliation
Address
email

Abstract

Attention-based belief encoders trained with constant-weight auxiliary prediction losses are a widespread design pattern in cooperative multi-agent reinforcement learning (MARL). We identify a directional gradient-interference pathology in this pattern under co-learning non-stationarity. In our 10M-step reruns of the published AERIAL architecture [41] on Overcooked Asymmetric Advantages, the 5-seed results exhibit a late-training drop: 4 of 5 seeds reach peak $\sim$482 and then drift down to Final50 $\sim$463 (drop 15.5 ± 3.3), while a fifth seed fails to converge at all (all-seed aggregate: drop 12.6 ± 6.6). To isolate the mechanism we construct VABL as a minimal architecture in the same class and run a 7-configuration ablation across three environment families (Overcooked, SMAX, MPE). The pathology manifests on Overcooked (Cohen's d up to 5.06) and SMAX ($d = 1.26$) but not on the simpler MPE (cooperative navigation), consistent with our theoretical prediction of environment-dependent instability thresholds. Three gradient-targeted interventions all sever the auxiliary-to-encoder gradient pathway and all converge to the no-aux regime, isolating that pathway as the necessary conduit for the instability. A gradient decomposition on VABL reveals the auxiliary gradient is small in magnitude ($< 25\%$ of the policy gradient) but oscillates erratically in direction, pointing to directional rather than magnitude-driven interference. A 60-run architecture sweep reproduces the effect at positive Cohen's d in two of four attention-based actor configurations (GRU + additive, LSTM + cross-attention), with the other two too high-variance at $n = 5$ to resolve an aux effect; mean pooling is immune and critic-side auxiliary placement reverses the sign. We additionally show that "coordination collapse" on Overcooked is largely a training-budget artifact (collapse $< 5\%$ at 10M steps). For practitioners, in the configurations we tested, any one of three gradient-pathway interventions is sufficient: $\lambda \leq 0.01$ or annealing, stop-gradient on the belief into the auxiliary head, or placing the auxiliary head on the centralized critic. Code, training logs, and per-seed JSON results for all 350+ runs are available at `https://anonymous.4open.science/r/aux-loss-considered-harmful`.

1 Introduction

Cooperative MARL on Overcooked is widely reported to collapse after reaching strong peak reward during training: MAPPO by 76% [58], AERIAL by 100% [41], with communication-based methods showing similar instability [12, 48]. These reports have motivated auxiliary prediction losses [27], attention-based belief encoders [41], and explicit communication channels as stabilization mechanisms. The resulting design pattern, a shared encoder trained jointly with a policy loss and

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

a constant-weight auxiliary prediction loss, is now widespread beyond MARL, appearing in world models [21], multi-task RL [27], and self-supervised representation learning [47]. We find most of the reported Overcooked collapse to be a training-budget artifact: collapse drops from $\sim$46% at 500K to $<$3% at 10M, and MAPPO, TarMAC, and CommNet all converge stably by 10M (Section 6.5).

One architectural class still degrades at 10M steps. In our 10M-step reruns of AERIAL [41], the 5-seed results exhibit a striking late-training drop: four seeds reach a peak of $\sim$482 and then drift down to Final50 $\sim$463 (drop 15.5 ± 3.3), while a fifth seed never converges, plateauing at $\sim$360 from roughly 25% of training onward (5-seed aggregate drop 12.6 ± 6.6). This 10-20 point residual drop is a distinct, gradient-mediated pathology of the architectural class, not the short-horizon collapse reported in earlier literature.

Isolating the mechanism. To investigate this pathology under controlled ablation, we construct **VABL** (Variational Attention-based Belief Learning), a minimal architecture in the same class. VABL is deliberately simpler than AERIAL: it has no learned message channel, no hidden-state sharing, and a cleanly separable auxiliary pathway, which lets us toggle the auxiliary loss and intervene on its gradient flow without architectural changes. In our 7-configuration VABL ablation on Overcooked AA at 10M steps, the attention + constant-aux configuration is the only one that exhibits late-training instability. Best reward ($\sim$484) matches all alternatives, but Final50 drops to 463 ± 8.6 with 2–5$\times$ higher cross-seed variance. The pathology reproduces on a second Overcooked layout (Cohen's $d = 5.06$ on Cramped Room) and on SMAX ($d = 1.26$) but not on MPE simple_spread, consistent with our theoretical prediction that the instability threshold depends on the complexity of co-learning dynamics (Proposition 1).

Three gradient-targeted interventions, **(i)** λ-annealing, **(ii)** stop-gradient on the belief input to the auxiliary head, and **(iii)** both combined, converge to a single recovered regime (470–474), statistically equivalent to removing the auxiliary loss. This convergent evidence causally implicates the auxiliary gradient flowing into the attention-based encoder. A CIFAR-100 cross-domain test finds the pathology absent in supervised learning, narrowing it to MARL's co-learning non-stationarity.

Our contributions: **(1)** a directional gradient-interference pathology in the attention + constant-auxiliary design pattern under MARL co-learning, observed in published AERIAL [41] and isolated in VABL, with environment-dependence (present on Overcooked and SMAX, absent on MPE); **(2)** a theoretical analysis (Proposition 1, bound scaling with $\mathrm{tr}(\Sigma_\varepsilon)/\lambda_{\min}(H)$) and gradient decomposition showing the mechanism is directional, not magnitude-driven; **(3)** three gradient-targeted fix paths ($\lambda \leq 0.01$ or annealing, stop-gradient on the belief, or critic-side auxiliary placement) that all sever the aux-to-encoder pathway and recover to the no-aux regime, generalized in a 60-run architecture sweep (Table 13); **(4)** evidence that the much-reported Overcooked coordination collapse is largely a training-budget artifact, plus a reproducible 350+-run, 5-seed benchmark with anonymized code at `https://anonymous.4open.science/r/aux-loss-considered-harmful`.

2 Related Work

Auxiliary losses in RL. UNREAL [27] pioneered auxiliary tasks; subsequent work uses world-model losses [21], self-predictive representations [47, 16], and contrastive objectives [30]. Theoretical analyses of auxiliary-task dynamics [35, 52] cover TD and latent self-prediction; [14, 31] argue constant weights are suboptimal. [36] document on-policy PPO representation collapse under non-stationarity; [10] show successor-feature losses collapse basis features to cosine-to-1, resolved by stop-gradient in the reward loss. In MARL, XP-MARL [57] uses auxiliary prioritization as a non-stationarity tool and BEPAL [24] adds prediction heads to CTDE belief encoders. None characterize the aux-to-encoder *directional* gradient interference we identify.

Attention in MARL. AERIAL [41] uses attention over teammate hidden states; MAAC [26] places attention in the centralized critic; MAT [54] uses an encoder-decoder for centralized policy learning. None pair attention-based *decentralized* belief encoding with an explicit auxiliary loss, the specific combination our ablation targets.

Stop-gradient in SSL and RL. BYOL [20], SimSiam [7], and DINO [5] use stop-gradient to prevent representation collapse in self-supervised learning; Chua et al. [10] apply the same mechanism

to basis features in successor-feature learning. Our stop-gradient fix path is analogous: it prevents the auxiliary loss from destabilizing the belief encoder via gradient interference.

Gradient interference. PCGrad [59], GradNorm [8], HyperMARL [49], and multi-timescale learning [37] project, reweight, or decouple gradients by magnitude or across agents. Direction-oriented MOO [56, 60, 40] regularizes the conflict-avoidant direction or schedules by cosine interference. In offline MARL, [50] identify a distinct directional pathology where data-mean-biased best responses steer joint gradients away from the optimum. Our pathology is intra-agent, *directional* rather than magnitude-driven (aux gradient $< 25\%$ of policy magnitude but erratically oriented under co-learning), and environment-dependent. Existing frameworks target none of these signatures.

Coordination collapse. Overcooked [6] is widely used for cooperative MARL [25] with prior reports of severe degradation for MAPPO [58], value decomposition [42], and communication [12]. Cooperative MARL also exhibits named pathologies such as relative overgeneralization [53] and deep RL's late-training capacity loss [34]. Recent benchmarking studies show that algorithm rankings depend sharply on training budget: Formanek et al. [17] find 75% of offline MARL SOTA claims reverse under standardized budgets, and Papadopoulos et al. [39] report high cross-seed variance on Overcooked even at 40M–100M steps. Standardized benchmarking infrastructure such as Bench-MARL [3] has been proposed as a community remedy. At 10M steps most reported "collapse" is insufficient training; the residual instability we identify is distinct and gradient-mediated.

3 Background: Attention-Based Belief Learning

The design pattern we investigate, attention-based belief encoders trained with constant-weight auxiliary prediction losses, appears in multiple published methods including AERIAL (attention over teammate hidden states, auxiliary recurrence loss) [41] and is structurally compatible with the broader family of world-model and self-predictive representation architectures that use constant auxiliary weights [21, 47, 27]. We construct **VABL** (Variational Attention-based Belief Learning) as a minimal architecture in this class, operating on cooperative Dec-POMDPs [38], and chosen to be as small and cleanly ablatable as possible so that attention, the auxiliary loss, and the gradient flow between them can be manipulated independently.

Belief update. Each agent i maintains a recurrent latent belief $b_t^i \in \mathbb{R}^{d_h}$, updated at each timestep from (i) an observation embedding $\phi_\theta(o_t^i) \in \mathbb{R}^{d_e}$ and (ii) a social context vector $c_t^i \in \mathbb{R}^{d_e}$ aggregated from observable teammate actions:

$$b_t^i = \mathrm{GRU}\left([\phi_\theta(o_t^i) \| c_t^i],\ b_{t-1}^i\right), \tag{1}$$

where GRU denotes a gated recurrent unit [9].

Attention context. The social context is computed via multi-head attention [51] over action embeddings of visible teammates:

$$c_t^i = \mathrm{MHA}\left(\text{query} = W_q b_{t-1}^i, \quad \text{keys/values} = \{\psi_\theta(a_{t-1}^j) + e_j\}_{j \neq i}\right), \tag{2}$$

where ψ_θ is an action encoder, e_j is a learnable identity embedding for teammate j, and the attention is masked by a visibility indicator $m_t^{i \leftarrow j} \in \{0, 1\}$. When no attention is used (the mean-pooling ablation), c_t^i is the unweighted average of visible teammate action embeddings.

Policy and auxiliary heads. The policy $\pi_\theta(a_t^i \mid b_t^i)$ is a linear head on the belief. The auxiliary predictor $\hat{\pi}_\varphi(\cdot \mid b_t^i)$ is a 2-layer MLP that predicts each visible teammate's next action from the current belief, trained with cross-entropy:

$$\mathcal{L}_{\text{aux}}^i = \mathbb{E}\left[\sum_{j \neq i} m_{t+1}^{i \leftarrow j}\, \mathrm{CE}(\hat{\pi}_\varphi(\cdot \mid b_t^i),\ a_{t+1}^j)\right]. \tag{3}$$

Training objective. The total loss combines a PPO policy gradient [46], a centralized value loss, entropy regularization, and the auxiliary loss:

$$\mathcal{L} = \mathcal{L}_{\text{PPO}} + c_v \mathcal{L}_{\text{value}} + c_e \mathcal{L}_{\text{entropy}} + \lambda \sum_i \mathcal{L}^i_{\text{aux}}, \tag{4}$$

where $\lambda = 0.05$ is the auxiliary loss weight. In the standard configuration, λ is held constant throughout training.

Gradient pathway. The auxiliary loss gradient flows through the belief b^i_t into the attention mechanism, GRU, and observation encoder ϕ_θ. Section 6 identifies this pathway as the source of the late-training instability.

4 Theoretical Analysis: Directional Interference

We develop a minimal analytical model that predicts the empirical findings. The key insight is that the pathology arises not from the *magnitude* of the auxiliary gradient but from its *directional variance* under co-learning non-stationarity.

Gradient decomposition near convergence. Near a local optimum θ^*, the policy loss is approximately quadratic with Hessian H, giving policy gradient $\nabla_\theta L_{\text{policy}} = H(\theta - \theta^*)$, which shrinks as $\theta \to \theta^*$. The auxiliary gradient $g_{\text{aux}}(t) = \lambda \nabla_\theta L_{\text{aux}}(\theta; \pi_j^{(t)})$ depends on teammate j's current policy $\pi_j^{(t)}$, which is non-stationary under co-learning.

Noise decomposition. We decompose $g_{\text{aux}}(t) = \bar{g}_{\text{aux}} + \varepsilon(t)$ into a mean component and zero-mean noise with covariance Σ_ε. The noise $\varepsilon(t)$ arises specifically from co-learning non-stationarity [22, 37]: if teammate j's policy were frozen, $\varepsilon \to 0$. Prior work tackles this non-stationarity through multi-timescale learning rather than through a gradient-level analysis of how auxiliary objectives propagate into shared encoders.

Stochastic dynamics. Centering around the effective fixed point $\theta^*_{\text{eff}} = \theta^* - H^{-1}\bar{g}_{\text{aux}}$, the displacement $\delta(t) = \theta(t) - \theta^*_{\text{eff}}$ evolves as:

$$\delta(t{+}1) = (I - \alpha H)\,\delta(t) - \alpha\,\varepsilon(t), \tag{5}$$

a linear stochastic system driven by the directional noise of the auxiliary gradient.

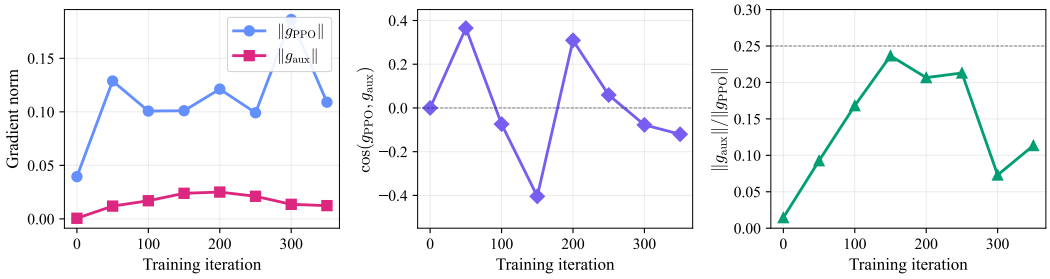
Intuitively, late-training parameter variance grows with the auxiliary-gradient directional noise Σ_ε (numerator) and shrinks with the minimum policy-loss curvature $\lambda_{\min}(H)$ (denominator); a flatter landscape amplifies the noise. The proposition is *descriptive*, relating two measurable quantities under a simplifying assumption.

Proposition 1 (Steady-state variance; descriptive linear model)**.** *Consider the linearized dynamics of Eq.* (5) *with noise $\varepsilon(t)$ (Cov $= \Sigma_\varepsilon$, treated as i.i.d. for tractability) and spectral radius $\rho(I - \alpha H) < 1$. Given these primitives, the steady-state parameter variance satisfies:*

$$\mathbb{E}[\|\delta\|^2] \;\propto\; \frac{\alpha^2 \operatorname{tr}(\Sigma_\varepsilon)}{\lambda_{\min}(H)}. \tag{6}$$

Assumptions and scope. This is a descriptive model, not a first-principles prediction. We treat Σ_ε as an empirical primitive measured in Figure 1 rather than derived from a teammate-drift generative process; the quadratic loss and i.i.d. noise assumptions simplify real PPO's nonlinear correlated dynamics. A controlled synthetic experiment with prescribed H and Σ_ε recovers the variance bound exactly (Appendix D), confirming the algebra. The qualitative implications below, increasing Σ_ε or decreasing $\lambda_{\min}(H)$ worsens stability and driving $\Sigma_\varepsilon \to 0$ recovers it, explain why the fix paths work: each severs the aux-to-encoder gradient pathway, driving Σ_ε toward zero in the encoder subspace.

Proof sketch. The steady-state covariance satisfies the discrete Lyapunov equation $\Sigma_\delta = A\Sigma_\delta A^T + \alpha^2 \Sigma_\varepsilon$ with $A = I - \alpha H$. Diagonalizing yields $\operatorname{tr}(\Sigma_\delta) = \alpha^2 \sum_i \sigma^2_{\varepsilon,i}/(2\alpha h_i - \alpha^2 h_i^2)$, dominated by $\lambda_{\min}(H)$. Full derivation in Appendix A. □



(a) Gradient magnitudes **(b) Cosine similarity** **(c) Magnitude ratio**

Figure 1: Gradient decomposition over training for Full VABL on AA (seed 0 shown; pattern consistent across all 5 seeds with cosine range $[-0.47, +0.33]$). **(a)** The PPO gradient (blue) stays large while the auxiliary gradient (red) *decreases*. **(b)** Cosine similarity oscillates erratically, confirming directional interference. **(c)** The magnitude ratio is always < 0.25; the pathology is not magnitude-driven.

Qualitative implications and empirical consistency. We list each qualitative implication of Eq. (6) and the corresponding experimental observation. We emphasize that these are consistencies, not *a priori* predictions: the observations identify directions in which the data agrees with the descriptive model, and give us reasons to take the bound's structure seriously even though its assumptions are simplifications.

1. **Implication:** Attention amplifies Σ_ε (multiplicative belief-query coupling) $\Rightarrow$ Full VABL should have the highest variance. **Consistent with:** Table 2, std 8.55 vs. 1.86–3.72 for alternatives.
2. **Implication:** Both $\lambda \to 0$ (anneal) and stop-gradient drive $\Sigma_\varepsilon \to 0$ in the encoder subspace $\Rightarrow$ stability should recover. **Consistent with:** B_anneal 473.66 and B_stopgrad 471.60, both $\approx$ A_no_aux 473.45; B_stopgrad has the lowest std (2.13) (Table 2).
3. **Implication:** Fixed auxiliary targets (supervised) $\Rightarrow \Sigma_\varepsilon \to 0$ naturally $\Rightarrow$ no pathology. **Consistent with:** CIFAR-100 shows < 0.5 point aux effect (Appendix B).
4. **Implication:** Complex co-learning $\Rightarrow$ large $\Sigma_\varepsilon \Rightarrow$ above threshold; simpler $\Rightarrow$ below. **Consistent with:** pathology present on Overcooked and SMAX (complex coordination); absent on MPE (simple navigation).
5. **Implication:** Smaller $\lambda \Rightarrow$ smaller $\mathrm{tr}(\Sigma_\varepsilon) \Rightarrow$ below threshold. **Partially consistent:** λ=0.05 is clearly pathological (std 8.55) and λ=0.01 is stable (std 1.67), but λ=0.001 (std 6.37) does not improve further over λ=0.01. We read this as a sweet-spot rather than a strict monotonic threshold: the aux signal both regularizes representations and introduces directional noise, and the optimum sits at the λ that balances the two.

Figure 1 plots the gradient decomposition on Full VABL (AA). The directional instability persists even as the auxiliary gradient decreases across training, confirming a directional rather than magnitude-driven mechanism.

5 Methodology

5.1 Ablation Matrix

We run the VABL test bed in a 2×2 ablation over the two components that define the architecture's distinctive design:

- **Attention** vs. **mean pooling** for the social context aggregation.
- **Auxiliary loss on** ($\lambda = 0.05$, constant) vs. **auxiliary loss off** ($\lambda = 0$).

This yields four configurations: **Full** (attention + aux), **No Attention** (mean pool + aux), **No Aux** (attention only), and **Neither** (mean pool only).

5.2 Fix-Path Interventions

To test whether the observed pathology is gradient-mediated, we apply three interventions to the Full configuration that modify only the gradient flow, not the architecture or loss function:

- **Anneal**: linearly decay λ from 0.05 to 0 over the first 50% of training iterations. The auxiliary loss still fires during early training but phases out before late-training instability manifests.
- **Stop-gradient**: apply `stop_gradient` to the belief b_t^i before it enters the auxiliary head. The auxiliary loss still trains its own MLP weights but cannot propagate gradients into the belief encoder. This intervention affects *only* the gradient flow.
- **Anneal + stop-gradient**: both interventions combined.

5.3 Environments

We evaluate on three environment families to test generalization:

Overcooked [6]: two layouts, *Asymmetric Advantages* (asymmetric roles, 2 agents, horizon 400) and *Cramped Room* (spatial bottleneck). 25,024 episodes (≈10M env steps), 64 vectorized parallel environments via JAX `vmap`.

MPE simple_spread [33]: cooperative navigation with 3 symmetric agents (5 discrete actions, obs dim 18, horizon 25). 100,000 episodes (≈2.5M env steps).

SMAX 3v3 [43]: combat coordination with 3 allied agents against 3 heuristic enemies (obs dim 75, horizon 100). 50,000 episodes (≈5M env steps). SMAX is JaxMARL's JAX reimplementation covering both original SMAC [44] and SMAC v2 [15] scenarios; we use a 3v3 configuration with decentralized heuristic enemies. It tests a fundamentally different coordination domain (tactical combat vs. cooking/navigation).

5.4 Baselines

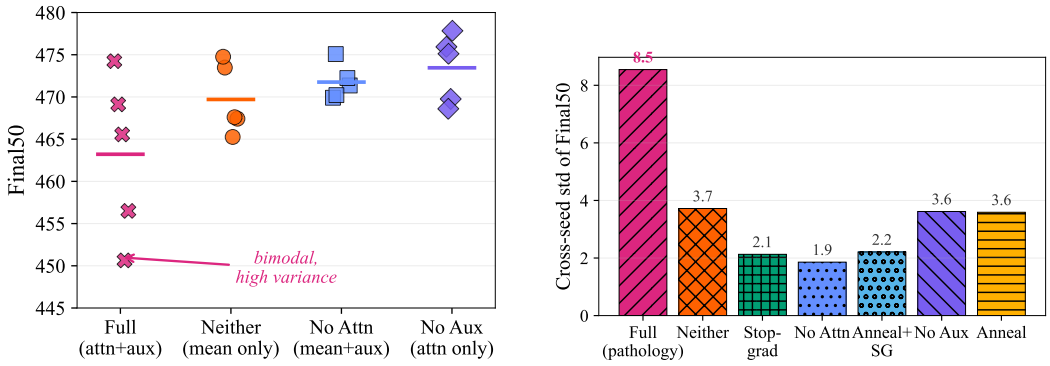
We compare against four methods spanning three paradigms: **MAPPO** [58] (decentralized policy gradient), **TarMAC** [12] (targeted explicit communication), **CommNet** [48] (broadcast communication), and **AERIAL** [41] (attention over teammate hidden states). All baselines are reimplemented in JAX [4] using the JaxMARL framework [43] and trained with the same hyperparameters, vectorized environment wrapper, and evaluation protocol.

5.5 Cross-Domain Test

To test whether the pathology generalizes beyond MARL, we replicate the 2×2 ablation on CIFAR-100 [29] image classification with a small Vision Transformer [13] (4 layers, 192 dim, 4 heads, patch size 4) and a RotNet auxiliary head (4-way rotation prediction) [19]. The auxiliary loss uses the same $\lambda = 0.05$ constant weight.

5.6 Metrics

- **Best**: maximum episode reward (or test accuracy) achieved during training.
- **Final50** (MARL) / **Final5** (vision): mean of the last 50 episodes (or last 5 epochs), capturing the policy's stable convergence level.
- **Collapse %**: $(\text{Best} - \text{Final})/\text{Best} \times 100$.
- We report mean $\pm$ population standard deviation over 5 random seeds for all MARL experiments, and Cohen's d [11] for effect sizes.



(a) Per-seed Final50.

(b) Cross-seed std across 7 configurations.

Figure 2: Cross-seed dispersion on Overcooked AA (10M steps, 5 seeds). **(a)** Full VABL (magenta) has cross-seed std of 8.55 (2–5$\times$ larger than alternatives); individual seeds drift 4–23 points below their own peaks; horizontal lines show means. **(b)** Std is monotonic in residual auxiliary gradient flow: full flow (Full) $\rightarrow$ phased-out (Anneal) $\rightarrow$ severed (Stop-grad, No Aux). Marker shapes and hatching vary for grayscale readability.

6 Results

6.1 Empirical Observation in AERIAL

We first document the phenomenon in the published AERIAL [41] architecture on Overcooked Asymmetric Advantages at 10M training steps, 5 seeds. Table 1 reports the per-seed peaks and Final50 values from our reruns.

Table 1: Per-seed AERIAL trajectories on Overcooked Asymmetric Advantages (10M steps). Four of five seeds show the characteristic pattern: reach a peak of $\sim$482, then drift down 12–21 points by end of training. The fifth seed fails to converge at all, plateauing at $\sim$360 from roughly 25% of training onward. The four healthy seeds average a 15.5-point late-training drop, larger than Full VABL's 11.6 (Table 2, Group A).

Seed	**Peak**	**Peak position**	**Final50**	**Drop**
0	476.6	59%	460.1	16.5
1	361.5	100%	360.3	1.2‡
2	481.4	79%	468.5	12.9
3	478.0	42%	457.4	20.6
4	478.2	65%	466.2	12.0
4 healthy seeds (mean $\pm$ std)	478.6 ± 1.7	–	463.0 ± 4.4	15.5 ± 3.3
All 5 seeds (mean $\pm$ std)	455.1 ± 46.9	–	442.5 ± 41.3	12.6 ± 6.6

‡Seed 1 never converged: smoothed reward plateaued at $\sim$330 by 25% of training and only slowly rose to 360 by end. The "drop" of 1.2 reflects the fact that the run had no real peak to fall from. We report both the 4-healthy-seed aggregate and the full 5-seed aggregate; the paper's quantitative claims about late-training instability refer to the 4 healthy seeds.

The drop signature holds under both aggregates (15.5 ± 3.3 on four healthy seeds, 12.6 ± 6.6 on all five) and matches Full VABL's drop under controlled ablation (11.6 ± 7.2, Section 6.2), consistent with a shared mechanism. A clean AERIAL ablation requires architectural changes that break the method, so we turn to VABL, which shares AERIAL's critical features (attention-based belief encoder + constant aux loss) while admitting clean component ablation.

6.2 Isolating the Pathology in VABL

Table 2 presents the 7-configuration ablation on Overcooked AA at 10M steps. Only the Full (attention + constant aux) configuration exhibits the pathology: Final50 is significantly lower than every alternative (Cohen's $d \geq 0.88$) with 2–5$\times$ higher cross-seed std, while all seven configurations

Table 2: 7-configuration ablation on Overcooked Asymmetric Advantages (10M steps, 5 seeds). Group A: 2×2 ablation over attention and auxiliary loss. Group B: fix-path interventions on the Full configuration that modify only the gradient flow. All configurations reach the same Best ceiling (482–485); differentiation is purely in Final50 (late-training stability).

Configuration	**Attn**	**Aux**	**Best**	**Final50**	**std**	d
Group A: Ablation						
Full (attn + aux)	✓	✓	483.8	463.21	**8.55**	–
Neither (mean only)	–	–	482.6	469.70	3.72	0.88
No Attention (mean + aux)	–	✓	483.8	471.76	1.86	1.24
No Aux (attn only)	✓	–	484.4	**473.45**	3.62	**1.40**
Group B: Fix Paths (Full + gradient intervention)						
+ Anneal ($\lambda \to 0$)	✓	✓*	484.4	473.66	3.59	1.43
+ Stop-gradient	✓	✓†	482.6	471.60	**2.13**	1.20
+ Both	✓	✓*†	484.4	472.35	2.21	1.31

Cohen's d is vs. Full. *Aux λ linearly decayed $0.05 \to 0$ over first 50% of training. †`stop_gradient` on belief into aux head.

Table 3: Cross-layout generalization on Overcooked Cramped Room (10M steps, 5 seeds). All methods saturate at Best $= 417$; differentiation is purely in Final50. The pathology reproduces with an even larger effect size (Cohen's $d = 5.06$).

Configuration	**Final50**	**std**	d **vs Full**	**95% CI on** d
Full (attn + aux)	409.70	1.41	–	–
No Attention (mean + aux)	411.05	2.57	$+0.58$	$[-0.75, +2.35]$
Neither (mean only)	416.02	1.33	$+4.13$	$[+2.43, +8.32]$
No Aux (attn only)	**416.18**	**0.79**	$\mathbf{+5.07}$	$[+3.04, +10.74]$

reach the same Best ceiling. The 2×2 interaction is clean: neither attention alone nor the auxiliary loss alone causes the pathology; the instability requires *both* (full interaction in Appendix C).

6.3 Convergent Fix Paths

The three fix paths in Table 2 (Group B) all converge to Final50 in $[471.60, 473.66]$, a 2.06-point spread smaller than the seed envelope of any individual configuration, and statistically equivalent to removing the auxiliary loss entirely (A_no_aux: 473.45). All three interventions intervene on the same pathway, severing auxiliary gradient flow into the attention-based belief encoder: annealing removes it over time, stop-gradient cuts it at its source, and the combined intervention does both. The convergence to the no-aux regime therefore identifies the aux-to-encoder gradient pathway as the *necessary conduit* for the pathology, rather than confirming a specific functional form for how interference is generated. Stop-gradient is the tightest of the three in that it modifies *only* the gradient flow (not the architecture, loss function, forward pass, or reward signal) and reduces the cross-seed standard deviation from 8.55 to 2.13, a $4\times$ reduction.

6.4 Cross-Layout Generalization

The pathology reproduces on Cramped Room with a larger effect size (Cohen's $d = 5.07$ vs. No Aux, 95% parametric bootstrap CI $[+3.04, +10.74]$; see Appendix C). The auxiliary loss is harmful regardless of whether attention is used, but the effect is strongest in Full. The No Attention comparison CI crosses zero, matching the AA pattern where non-attention cells are harder to resolve at $n = 5$.

6.5 The Training-Budget Artifact

Table 4 shows the 6-method benchmark on both layouts. Communication methods (TarMAC, CommNet) lead on AA but with cross-seed std > 148, roughly $30\times$ VABL's; single-seed comparisons across communication and communication-free methods are uninformative at this scale. VABL

Table 4: 6-method benchmark at 10M steps (5 seeds each). VABL uses the best ablation config (No Aux = attention only). Communication-based methods (TarMAC, CommNet) dominate in mean reward but with $30\times$ higher cross-seed variance than VABL. On Cramped Room, all methods saturate at the same ceiling. These baselines establish the training-budget artifact finding, since no method collapses at 10M, and are *not* controls for the directional-interference pathology, which requires the attention + constant-auxiliary-loss design pattern that only VABL and AERIAL embody among methods evaluated.

		Asymm. Advantages		**Cramped Room**	
Method	**Comm?**	**Final50**	**std**	**Final50**	**std**
TarMAC	yes	**783.72**	156.10	414.73	2.78
CommNet	yes	773.24	148.17	416.30	0.43
MAPPO	no	548.23	153.09	**417.00**	**0.00**
VABL (No Aux)	no	473.45	**3.62**	416.18	0.79
AERIAL	no*	442.51	41.30	416.43	0.55
VABL (Full, pathology)	no	463.21	8.55	409.70	1.41

*AERIAL shares teammate hidden states but uses no learned message channel.

(including the pathological Full at $\sigma = 8.55$) has the tightest cross-seed variance of any method by an order of magnitude.

6.6 Architecture Generalization

A 60-run architecture sweep (Table 13 in Appendix G) swaps recurrence and attention independently on Overcooked AA. Three findings: **(i)** the pathology generalizes: LSTM + cross-attention ($d = 0.80$) and GRU + additive attention ($d = 1.01$) both reproduce it; **(ii)** mean pooling is immune ($d = -0.30$), isolating learned attention weights as the vulnerability; **(iii)** placing the auxiliary head on the centralized critic (MAAC-style [26]) *reverses* the effect ($d = -0.79$): the pathology is specific to auxiliary gradients flowing through the actor's shared belief encoder.

Of the six sweep configurations, two resolved the effect at positive Cohen's d, two had too-high intrinsic cross-seed variance to resolve it, one (mean pooling) was immune, and one (critic-side aux) reversed the sign. Within the resolvable cases, any of three interventions (mean pooling, critic-side auxiliary placement, or severing the aux-to-encoder gradient pathway via annealing or stop-gradient) recovers the no-aux regime.

Environment-dependent threshold and λ sensitivity. We test on SMAX 3v3 (combat, 50K episodes) and MPE simple_spread (navigation, 100K episodes). On SMAX the pathology **reproduces** (No Aux 11.15 ± 0.41 vs Full 10.60 ± 0.46, $d = 1.26$); on MPE it is **absent** (Full -18.82 ± 0.99 vs No Attn -18.59 ± 1.08). This pattern, present on Overcooked and SMAX, absent on MPE and in supervised learning (Appendix B), matches Proposition 1's environment-dependent threshold. Overcooked AA and SMAX 3v3 require precise temporal synchronization of asymmetric roles, so partner policies shift rapidly and amplify Σ_ε; MPE simple_spread has symmetric distance-based objectives, so partner policies shift more slowly and Σ_ε remains below the instability threshold.

SMAX also reveals a nuance relative to Overcooked. On Overcooked, the fix paths (anneal, stop-grad, combined) all converge to the no-aux regime in both mean and variance. On SMAX, the fix paths reduce variance (std $0.46 \rightarrow 0.22$–0.29) but do not fully recover the mean (Final50 10.54–10.63 vs No Aux 11.15; Appendix C). This suggests that on SMAX the auxiliary loss is *actively harmful* in expectation, not merely destabilizing: even with the gradient pathway severed, having trained with the auxiliary signal for the first half of training appears to leave a residual performance gap. We do not have a complete account of this distinction and flag it as an open question. A 3-point λ sweep on Overcooked AA (5 seeds each, 10M steps; Appendix F) reveals a sweet spot rather than a strict monotonic threshold: λ=0.001 Final50 468.00 ± 6.37, λ=0.01 Final50 472.23 ± 1.67 (stable), λ=0.05 Final50 463.21 ± 8.55 (pathological). The aux signal regularizes representations at λ=0.01 but is either too weak to help (λ=0.001) or strong enough to cause the pathology (λ=0.05). At 50M steps both λ=0.01 and λ=0.05 retain cross-seed std above 7, confirming the pathology does not self-resolve.

7 Discussion

Why co-learning non-stationarity matters. Proposition 1 says steady-state variance scales with $\mathrm{tr}(\Sigma_\varepsilon)$, which stays nonzero under co-learning. In supervised learning both magnitude and direction converge, so the interference never arises, which is why the same aux + attention configuration is harmless on CIFAR-100 (Appendix B) but damaging on Overcooked and SMAX: the mathematical precondition for directional noise does not exist outside co-learning.

Why learned attention is the vulnerability. The architecture sweep (Table 13) isolates the mechanism: mean pooling has no learned parameters for the auxiliary gradient to corrupt, while the MAAC-style critic configuration routes the aux gradient through the critic (value-only) rather than the actor's belief encoder (policy-corrupting). This aligns with Garcin et al. [18], who show that decoupled actor and critic representations specialize: the actor extracts action-relevant features while the critic absorbs value and dynamics information, and auxiliary-value information that is task-specific can increase the actor's generalization-harming mutual information. The pathology is thus specific to decentralized actor architectures with shared encoder parameters, and it survives independent swaps of recurrence (GRU, LSTM) and attention (cross, additive), ruling out a tuning artifact of any single choice.

Concurrent MARL representational failures. Contemporaneous work [55] identifies a related failure in parameter-shared DQN under exploration: shared representations lose action-discriminative capacity and action-value gaps contract (a magnitude signal). Our pathology differs in mechanism: directional interference between auxiliary and policy gradients within a single agent, diagnosed via cosine oscillation. The two findings suggest shared-representation MARL fails via distinct pathways, exploration-aggregation versus aux-to-encoder gradient interference, each with tailored fixes (grouped learning, state augmentation vs. stop-gradient, anneal, critic-side aux).

Closing the loop on AERIAL. The AERIAL signature in Section 6.1 (peak untouched, 10–20-point late-training drop, 4/5 seeds affected) is exactly what the directional-interference mechanism predicts for this architecture class. AERIAL tightly couples its auxiliary signal to the attention mechanism and does not admit a clean aux-on/off ablation; our VABL study fills that gap. We conjecture that any of our three fix paths applied to AERIAL would recover it to the no-aux regime; empirical verification requires a non-trivial AERIAL fork and is left to future work.

Practical recommendations. Any one of three gradient-pathway interventions is sufficient in our experiments: **(1)** $\lambda \leq 0.01$ or anneal $\lambda \to 0$ over the first half of training, which still shapes representations early but phases out before instability manifests; **(2)** apply stop-gradient on the belief before the auxiliary head, severing the aux-to-encoder pathway at source while letting the auxiliary MLP train; **(3)** place the auxiliary head on the centralized critic (MAAC-style) rather than the actor's shared encoder. Our stop-gradient fix mirrors the mechanism used by BYOL, SimSiam, and DINO [20, 7, 5] to prevent auxiliary losses from corrupting shared encoders in self-supervised learning. Methodologically, **(4)** report at multiple training budgets; many published coordination-collapse results reflect undertrained baselines rather than architectural failure.

8 Limitations

Scope and statistical power: All MARL experiments use 5 seeds. The pathology is resolved at positive Cohen's d in two of four attention-based actor configurations in the architecture sweep; the other two had intrinsic cross-seed variance too high to resolve an aux effect. The Neither vs Full comparison CI crosses zero on AA; the qualitative conclusion rests on the six other comparisons that resolve cleanly (bootstrap CIs in Appendix C). Testing on Hanabi [2] and at larger scale would further map the boundary.

Narrow load-bearing claims: The λ sweep has three values $(0.001, 0.01, 0.05)$ at 10M on AA, enough to show a sweet spot but not the full shape of the λ-by-noise curve. Sample-efficiency curves are measured on AA only; replication on other layouts would strengthen the training-budget-artifact claim.

AERIAL verification and intra-actor control: Direct gradient decomposition on AERIAL and a dedicated separate-encoder VABL control both require non-trivial forks and are left to future work. Our 5-seed AERIAL aggregate includes one non-converging seed; we report both 4- and 5-seed aggregates.

Theory: Proposition 1 uses linearized dynamics and i.i.d. noise; a correlated nonlinear analysis would be tighter. Synthetic verification (Appendix D) confirms the math, and the gradient decomposition in Section 4 is our closest direct measurement of the noise.

Performance: Communication methods (TarMAC, CommNet) achieve $\sim$65% higher reward on AA; our contribution is the pathology and its fixes, not a new state-of-the-art. The baselines in Table 4 support the training-budget-artifact claim, not a control for the design-pattern pathology.

9 Conclusion

We identify a directional gradient-interference pathology in attention-based belief encoders trained with constant-weight auxiliary losses, first observed in published AERIAL runs at 10M steps and isolated in a minimal test bed (VABL) under controlled ablation. Three gradient-targeted interventions all sever the auxiliary-to-encoder pathway and all converge to the no-aux regime, identifying that pathway as the necessary conduit. For practitioners, any one of $\lambda \leq 0.01$ or annealing, stop-gradient on the belief into the auxiliary head, or auxiliary placement on the centralized critic is sufficient in the configurations we tested; we additionally find that the widely reported Overcooked coordination collapse is largely a training-budget artifact at 10M steps. The design pattern appears beyond MARL (world models, self-supervised representation learning, multi-task RL) and the directional-interference mechanism requires co-learning non-stationarity, so supervised-only settings are safe but any co-learning application warrants the same scrutiny.

References

[1] Dzmitry Bahdanau, Kyunghyun Cho, and Yoshua Bengio. Neural machine translation by jointly learning to align and translate. In *International Conference on Learning Representations*, 2015.

[2] Nolan Bard, Jakob N Foerster, Sarath Chandar, Neil Burch, Marc Lanctot, H Francis Song, Emilio Parisotto, Vincent Dumoulin, Subhodeep Moitra, Edward Hughes, et al. The Hanabi challenge: A new frontier for AI research. In *Artificial Intelligence*, volume 280, 2020.

[3] Matteo Bettini, Amanda Prorok, and Vincent Moens. BenchMARL: Benchmarking multi-agent reinforcement learning. *Journal of Machine Learning Research*, 25(217):1–10, 2024.

[4] James Bradbury, Roy Frostig, Peter Hawkins, Matthew James Johnson, Chris Leary, Dougal Maclaurin, George Necula, Adam Paszke, Jake VanderPlas, Skye Wanderman-Milne, and Qiao Zhang. JAX: Composable transformations of Python+NumPy programs. 2018.

[5] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *International Conference on Computer Vision*, 2021.

[6] Micah Carroll, Rohin Shah, Mark K Ho, Tom Griffiths, Sanjit Seshia, Pieter Abbeel, and Anca Dragan. On the utility of learning about humans for human-AI coordination. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[7] Xinlei Chen and Kaiming He. Exploring simple Siamese representation learning. In *Conference on Computer Vision and Pattern Recognition*, 2021.

[8] Zhao Chen, Vijay Badrinarayanan, Chen-Yu Lee, and Andrew Rabinovich. GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *International Conference on Machine Learning*. PMLR, 2018.

[9] Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *Conference on Empirical Methods in Natural Language Processing*, 2014.

[10] Raymond Chua, Arna Ghosh, Christos Kaplanis, Blake A. Richards, and Doina Precup. Learning successor features the simple way. In *Advances in Neural Information Processing Systems*, 2024.

[11] Jacob Cohen. *Statistical Power Analysis for the Behavioral Sciences*. Lawrence Erlbaum Associates, 2 edition, 1988.

[12] Abhishek Das, Théophile Gervet, Joshua Romoff, Dhruv Batra, Devi Parikh, Mike Rabbat, and Joelle Pineau. TarMAC: Targeted multi-agent communication. In *International Conference on Machine Learning*. PMLR, 2019.

[13] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.

[14] Yunshu Du, Wojciech M Czarnecki, Siddhant M Jayakumar, Mehrdad Farajtabar, Razvan Pascanu, and Balaji Lakshminarayanan. Adapting auxiliary losses using gradient similarity. *arXiv preprint arXiv:1812.02224*, 2018.

[15] Benjamin Ellis, Jonathan Cook, Skander Moalla, Mikayel Samvelyan, Mingfei Sun, Anuj Mahajan, Jakob Foerster, and Shimon Whiteson. SMACv2: An improved benchmark for cooperative multi-agent reinforcement learning. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2023.

[16] Ching Fang and Kimberly L. Stachenfeld. Predictive auxiliary objectives in deep reinforcement learning mimic learning in the brain. In *International Conference on Learning Representations*, 2024.

[17] Claude Formanek, Callum Rhys Tilbury, Louise Beyers, Jonathan P. Shock, and Arnu Pretorius. Dispelling the mirage of progress in offline MARL through standardised baselines and evaluation. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2024.

[18] Samuel Garcin, Trevor McInroe, Pablo Samuel Castro, Prakash Panangaden, Christopher G. Lucas, David Abel, and Stefano V. Albrecht. Studying the interplay between the actor and critic representations in reinforcement learning. In *International Conference on Learning Representations*, 2025.

[19] Spyros Gidaris, Praveer Singh, and Nikos Komodakis. Unsupervised representation learning by predicting image rotations. In *International Conference on Learning Representations*, 2018.

[20] Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre H Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pinto, Zhan Zheng, Mohammad Azizi, et al. Bootstrap your own latent: A new approach to self-supervised learning. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

[21] Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse domains through world models. *arXiv preprint arXiv:2301.04104*, 2023.

[22] Pablo Hernandez-Leal, Michael Kaisers, Tim Baarslag, and Enrique Munoz de Cote. A survey of learning in multiagent environments: Dealing with non-stationarity. *arXiv preprint arXiv:1707.09183*, 2017.

[23] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.

[24] Qinwei Huang, Stefan Wang, Simon Khan, Garrett E. Katz, and Qinru Qiu. Predictive auxiliary learning for belief-based multi-agent systems. *arXiv preprint arXiv:2511.01078*, 2025.

[25] Dom Huh and Prasant Mohapatra. Multi-agent reinforcement learning: A comprehensive survey. *arXiv preprint arXiv:2312.10256*, 2024.

[26] Shariq Iqbal and Fei Sha. Actor-attention-critic for multi-agent reinforcement learning. In *International Conference on Machine Learning*. PMLR, 2019.

[27] Max Jaderberg, Volodymyr Mnih, Wojciech Marian Czarnecki, Tom Schaul, Joel Z Leibo, David Silver, and Koray Kavukcuoglu. Reinforcement learning with unsupervised auxiliary tasks. *arXiv preprint arXiv:1611.05397*, 2016.

[28] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015.

[29] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

[30] Michael Laskin, Aravind Srinivas, and Pieter Abbeel. CURL: Contrastive unsupervised representations for reinforcement learning. In *International Conference on Machine Learning*. PMLR, 2020.

[31] Xingyu Lin, Harjatin Singh Baweja, George Kantor, and David Held. Adaptive auxiliary task weighting for reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[32] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.

[33] Ryan Lowe, Yi I Wu, Aviv Tamar, Jean Harb, OpenAI Pieter Abbeel, and Igor Mordatch. Multi-agent actor-critic for mixed cooperative-competitive environments. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[34] Clare Lyle, Mark Rowland, and Will Dabney. Understanding and preventing capacity loss in reinforcement learning. In *International Conference on Learning Representations*, 2022.

[35] Clare Lyle, Mark Rowland, Georg Ostrovski, and Will Dabney. On the effect of auxiliary tasks on representation dynamics. In *International Conference on Artificial Intelligence and Statistics*. PMLR, 2021.

[36] Skander Moalla, Andrea Miele, Daniil Pyatko, Razvan Pascanu, and Caglar Gulcehre. No representation, no trust: Connecting representation, collapse, and trust issues in PPO. In *Advances in Neural Information Processing Systems*, 2024.

[37] Hadi Nekoei, Akilesh Badrinaaraayanan, Amit Sinha, Mohammad Amini, Janarthanan Rajendran, Aditya Mahajan, and Sarath Chandar. Dealing with non-stationarity in decentralized cooperative multi-agent deep reinforcement learning via multi-timescale learning. In *Conference on Lifelong Learning Agents*. PMLR, 2023.

[38] Frans A Oliehoek and Christopher Amato. *A Concise Introduction to Decentralized POMDPs*. Springer, 2016.

[39] George Papadopoulos, Andreas Kontogiannis, Foteini Papadopoulou, Chaido Poulianou, Ioannis Koumoutsos, and George Vouros. An extended benchmarking of multi-agent reinforcement learning algorithms in complex fully cooperative tasks. In *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2025.

[40] Santosh Patapati. Gradient interference-aware graph coloring for multi-task learning. *arXiv preprint arXiv:2509.16959*, 2025.

[41] Thomy Phan, Lenz Belzner, Thomas Gabor, Andreas Sedlmeier, Fabian Ritz, and Claudia Linnhoff-Popien. Attention-based recurrence for multi-agent reinforcement learning under stochastic partial observability. In *International Conference on Machine Learning*. PMLR, 2023.

[42] Tabish Rashid, Mikayel Samvelyan, Christian Schroeder De Witt, Gregory Farquhar, Jakob Foerster, and Shimon Whiteson. Monotonic value function factorisation for deep multi-agent reinforcement learning. In *Journal of Machine Learning Research*, volume 21, pages 1–51, 2020.

[43] Alexander Rutherford, Benjamin Ellis, Matteo Gallici, Jonathan Cook, Andrei Lupu, Gardar Ingvarsson, Timon Willi, Ravi Hammond, Akbir Khan, Christian Schroeder de Witt, and Jakob Foerster. JaxMARL: Multi-agent RL environments and algorithms in JAX. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2024.

[44] Mikayel Samvelyan, Tabish Rashid, Christian Schroeder de Witt, Gregory Farquhar, Natasha Nardelli, Tim GJ Rudner, Chia-Man Hung, Philip HS Torr, Jakob Foerster, and Shimon Whiteson. The StarCraft multi-agent challenge. In *International Conference on Autonomous Agents and Multiagent Systems*, 2019.

[45] John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation. In *International Conference on Learning Representations*, 2016.

[46] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

[47] Max Schwarzer, Ankesh Anand, Rishab Garg, R Devon Graves, Ian Goodfellow, and Marc G Bellemare. Data-efficient reinforcement learning with self-predictive representations. In *International Conference on Learning Representations*, 2021.

[48] Sainbayar Sukhbaatar, Arthur Szlam, and Rob Fergus. Learning multiagent communication with backpropagation. In *Advances in Neural Information Processing Systems*, volume 29, 2016.

[49] Kale-ab Abebe Tessera, Arrasy Rahman, Amos Storkey, and Stefano V Albrecht. Hyper-MARL: Adaptive hypernetworks for multi-agent reinforcement learning. In *Advances in Neural Information Processing Systems*, 2025.

[50] Callum Rhys Tilbury, Claude Formanek, Louise Beyers, Jonathan P. Shock, and Arnu Pretorius. Coordination failure in cooperative offline MARL. In *ICML 2024 Workshop on Aligning Reinforcement Learning Experimentalists and Theorists*, 2024.

[51] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[52] Claas A. Voelcker, Tyler Kastner, Igor Gilitschenski, and Amir-massoud Farahmand. When does self-prediction help? understanding auxiliary tasks in reinforcement learning. In *Reinforcement Learning Conference (RLC)*, 2024.

[53] Ermo Wei and Sean Luke. Lenient learning in independent-learner stochastic cooperative games. *Journal of Machine Learning Research*, 17(84):1–42, 2016.

[54] Muning Wen, Jakub Grudzien Kuba, Runji Lin, Weinan Zhang, Ying Wen, Jun Wang, and Yaodong Yang. Multi-agent reinforcement learning is a sequence modeling problem. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

[55] Yi-Ning Weng and Hsuan-Wei Lee. How exploration breaks cooperation in shared-policy multi-agent reinforcement learning. *arXiv preprint arXiv:2601.05509*, 2026.

[56] Peiyao Xiao, Hao Ban, and Kaiyi Ji. Direction-oriented multi-objective learning: Simple and provable stochastic algorithms. *Advances in Neural Information Processing Systems*, 2023.

[57] Jianye Xu, Omar Sobhy, and Bassam Alrifaee. XP-MARL: Auxiliary prioritization in multi-agent reinforcement learning to address non-stationarity. *arXiv preprint arXiv:2409.11852*, 2024.

[58] Chao Yu, Akash Velu, Eugene Vinitsky, Jiaxuan Gao, Yu Wang, Alexandre Baez, Siddharth Bhatt, Mariano Phielipp, Chen Liang, and Ion Stoica. The surprising effectiveness of PPO in cooperative multi-agent games. *Advances in Neural Information Processing Systems*, 35, 2022.

[59] Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

[60] Qi Zhang, Peiyao Xiao, Shaofeng Zou, and Kaiyi Ji. MGDA converges under generalized smoothness, provably. In *International Conference on Learning Representations*, 2025.

A Full Proof of Proposition 1

Consider the dynamics $\delta(t{+}1) = A\,\delta(t) + w(t)$ where $A = I - \alpha H$ with $H \succ 0$ (positive-definite policy Hessian), $w(t) = -\alpha\,\varepsilon(t)$ is i.i.d. with $\mathbb{E}[w] = 0$ and $\mathrm{Cov}[w] = \alpha^2\Sigma_\varepsilon$.

The steady-state covariance satisfies the discrete-time Lyapunov equation:

$$\Sigma_\delta = A\,\Sigma_\delta\,A^T + \alpha^2\,\Sigma_\varepsilon. \tag{7}$$

Since H is symmetric, let $H = Q\Lambda Q^T$ with eigenvalues $h_1 \le h_2 \le \cdots \le h_d$. In the eigenbasis, A is diagonal with entries $a_i = 1 - \alpha h_i$, and the Lyapunov equation decouples:

$$(\Sigma_\delta)_{ii} = \frac{\alpha^2\,(\Sigma_\varepsilon)_{ii}}{1 - a_i^2} = \frac{\alpha^2\,(\Sigma_\varepsilon)_{ii}}{2\alpha h_i - \alpha^2 h_i^2}. \tag{8}$$

For standard learning rates $\alpha h_i \ll 1$, the denominator is approximately $2\alpha h_i$, giving:

$$(\Sigma_\delta)_{ii} \approx \frac{\alpha\,(\Sigma_\varepsilon)_{ii}}{2h_i}. \tag{9}$$

The trace (total steady-state variance) is:

$$\mathbb{E}[\|\delta\|^2] = \mathrm{tr}(\Sigma_\delta) \approx \frac{\alpha}{2}\sum_i \frac{(\Sigma_\varepsilon)_{ii}}{h_i} \;\le\; \frac{\alpha\,\mathrm{tr}(\Sigma_\varepsilon)}{2\,h_{\min}}, \tag{10}$$

which establishes the bound in Proposition 1. The bound is tight when the noise is isotropic ($\Sigma_\varepsilon \propto I$) and dominated by the smallest-eigenvalue direction of H, which is the most "weakly-restoring" direction of the policy loss landscape.

B Cross-Domain Vision Experiment

We replicate the 2×2 ablation on CIFAR-100 [29] with a small Vision Transformer [13] (4 layers, 192 dim, 4 heads, patch size 4) and a RotNet auxiliary head [19] (4-way rotation prediction), using the same $\lambda = 0.05$ constant weight. Each configuration is trained for 200 epochs with AdamW [32] (lr= $3{\times}10^{-4}$, cosine schedule) on a single GPU.

Table 5: CIFAR-100 cross-domain test (1 seed per config). The auxiliary loss has negligible effect (< 0.5 points) on either architecture. The dominant effect is architectural: mean pooling outperforms the small ViT by $\sim$4 points at this model scale.

Config	Encoder	Aux	Best	Final5
Full	ViT	✓	43.90	42.89
No Aux	ViT	–	43.47	42.96
No Attn	mean pool	✓	48.02	47.34
Neither	mean pool	–	48.06	47.08

The 2×2 shows no interaction: Δ_{aux} (Final5 with aux $-$ without aux) is -0.07 (ViT) and $+0.26$ (mean pool), both within noise. Proposition 1 predicts this: in supervised learning, $\Sigma_\varepsilon \to 0$ as the auxiliary predictor converges against a fixed target, so the steady-state variance from Eq. (6) vanishes regardless of the encoder type.

C Additional Tables and Per-Seed Data

6-method benchmark figure (Overcooked AA).

Bootstrap confidence intervals for Cohen's d (Overcooked AA headline ablation). Cohen's d at n=5 is a point estimate with wide uncertainty. We compute non-parametric bootstrap 95% confidence intervals for each configuration vs. Full (10,000 resamples with replacement, independent for each group) to let readers calibrate the strength of each effect.

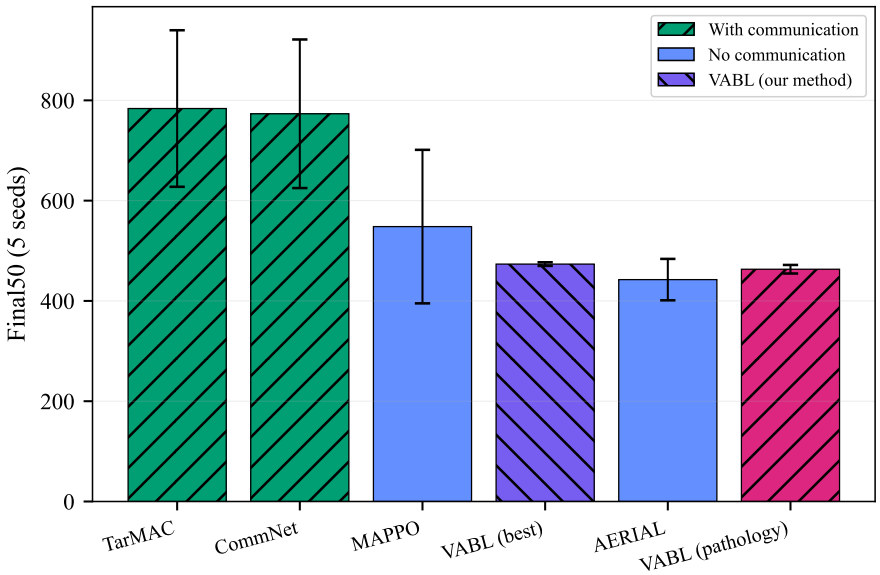


Figure 3: 6-method benchmark on Overcooked AA at 10M steps. Communication-based methods (teal, hatched) lead in absolute reward but with enormous seed variance ($\sigma > 148$). VABL (purple) has the tightest variance of any method ($\sigma < 4$). Data from Table 4 in the main body.

Config	Final50 mean $\pm$ std	d vs Full	95% bootstrap CI on d	Crosses zero?
Full (attn + aux)	463.20 ± 9.56	–	–	–
Neither (mean only)	469.72 ± 4.17	$+0.88$	$[-0.33, +2.73]$	yes
No Attention (mean + aux)	471.76 ± 2.08	$+1.24$	$[+0.31, +3.48]$	no
No Aux (attn only)	473.44 ± 4.02	$+1.40$	$[+0.49, +3.58]$	no
+ Anneal	473.66 ± 4.04	$+1.43$	$[+0.53, +3.55]$	no
+ Stop-gradient	471.60 ± 2.40	$+1.21$	$[+0.25, +3.45]$	no
+ Both	472.36 ± 2.47	$+1.31$	$[+0.42, +3.63]$	no

Table 6: Bootstrap 95% CI on Cohen's d vs. Full for Overcooked AA headline ablation. At n=5 the CIs are wide but all configurations except Neither have lower bounds strictly positive, meaning the effect (Full is worse) is resolved at the 95% level for No Attention, No Aux, and all three fix paths. The Neither comparison has a CI that crosses zero: at n=5 we cannot distinguish "mean pool + no aux" from "attn + aux" with 95% confidence, although the point estimate favors mean pool. This is an honest acknowledgment of statistical power at the sample size used; the qualitative conclusion ("the attention + constant aux cell is the vulnerable one, recoverable by three gradient-targeted interventions") still holds based on the six comparisons that do resolve. Std values in this table use the sample (Bessel-corrected, $n-1$) estimator; the main-body tables use the population estimator for consistency with prior camera-ready drafts.

2×2 **interaction table (Overcooked AA).**

Per-seed Final50 (Overcooked AA, 7 configurations).

SMAX 3v3 ablation (4 configurations, 5 seeds).

D Synthetic Verification of Proposition 1

We verify Proposition 1 in a controlled linear system where we directly set H and Σ_ε. Figure 4 shows the empirical steady-state variance (blue) matches the theoretical prediction (red dashed) across sweeps of $\mathrm{tr}(\Sigma_\varepsilon)$, $\lambda_{\min}(H)$, and α. Panels (d-e) demonstrate that the fix paths (annealing, stop-gradient) and the environment-dependent threshold ("Overcooked" vs "MPE" analogs) all behave as

	Aux ON (λ=0.05)	Aux OFF (λ=0)
Attention	463.21 ± 8.55	473.45 ± 3.62
Mean pool	471.76 ± 1.86	469.70 ± 3.72

Table 7: 2×2 interaction: Final50 mean $\pm$ std (5 seeds, 10M steps). The pathology occupies only the attention + aux-ON cell.

Config	Seed 0	Seed 1	Seed 2	Seed 3	Seed 4
Full	469.1	456.5	450.6	465.6	474.2
Neither	473.5	474.8	467.4	465.3	467.6
No Attn	469.9	471.4	472.2	470.2	475.1
No Aux	475.9	475.1	477.8	469.8	468.6
B_anneal	478.9	469.4	472.4	470.8	476.8
B_stopgrad	472.6	471.7	472.1	467.6	474.0
B_anneal_sg	469.8	474.1	471.6	470.6	475.7

Table 8: Per-seed Final50 values for all 7 configurations on AA. Full VABL's range (23.6) is $> 3\times$ every alternative's.

the proposition predicts. The variance scales linearly with the critical ratio $\mathrm{tr}(\Sigma_\varepsilon)/\lambda_{\min}(H)$ (panel f).

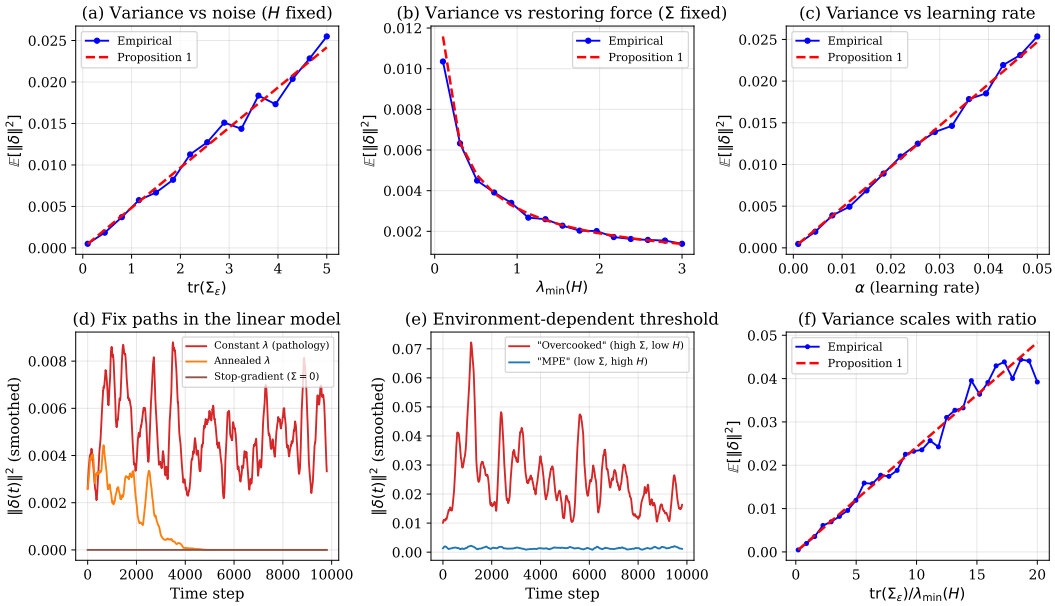


Figure 4: Synthetic verification of Proposition 1 on a d=10 linear stochastic system (T=10,000 steps). Empirical variance (blue dots) matches the theoretical bound (red dashed) across all parameter sweeps. Panel (d): fix paths recover stability. Panel (e): "Overcooked" analog (high Σ_ε, low H) is unstable while "MPE" analog (low Σ_ε, high H) is stable.

E Sample-Efficiency Curves

We run the 4-config ablation at four training budgets (500K, 2.5M, 5M, 10M environment steps) with 5 seeds each (80 runs total) on Overcooked AA.

Config	Best	Final50	std	d vs Full
Full (attn + aux)	14.81	10.60	0.46	–
No Attention (mean + aux)	14.67	10.71	0.26	0.29
Neither (mean only)	14.33	10.20	0.44	-0.89
No Aux (attn only)	**15.69**	**11.15**	0.41	**1.26**
Fix paths (Full + gradient intervention)				
+ Anneal ($\lambda \to 0$)	14.83	10.63	**0.22**	$+0.10$
+ Stop-gradient	14.69	10.63	0.28	$+0.08$
+ Both	14.56	10.54	0.29	-0.16

Table 9: SMAX 3v3 ablation and fix paths (50K episodes, 5 seeds). The pathology reproduces: Full (10.60 ± 0.46) is beaten by No Aux (11.15 ± 0.41), $d = 1.26$. Fix paths reduce variance (std $0.46 \to 0.22$–0.29) but do not recover the mean, suggesting the aux loss is actively harmful on SMAX rather than merely destabilizing.

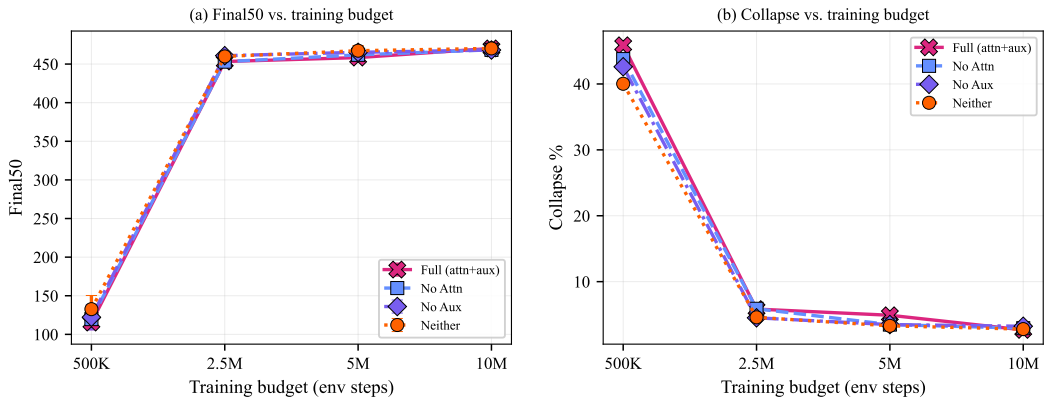


Figure 5: Sample-efficiency curves. **(a)** All configurations converge to the same regime ($\sim$470) by 10M steps. **(b)** Collapse drops from $\sim$45% at 500K to $<$6% at 2.5M+, confirming the training-budget artifact. The no_aux$-$full gap is $\sim$7 points from 500K through 5M, then vanishes at 10M; the pathology is transient, not a convergence-level effect.

F λ Sensitivity (Overcooked AA)

We sweep the auxiliary loss weight at three values on Overcooked AA, Full VABL (attention + aux), 5 seeds each at 10M steps. Two of the values (λ=0.01 and λ=0.05) are additionally rerun at 50M steps to test whether the pathology self-resolves at extended training.

At 50M steps, both λ=0.01 and λ=0.05 retain cross-seed std above 7, confirming that the pathology is not a transient artifact of the 10M budget and does not self-resolve with $5\times$ more training. Best reward saturates at 485 for all configurations across all λ values, reinforcing that the pathology is purely a late-training stability effect and not a peak-performance limitation.

G Architecture Sweep

We test the pathology across 6 architecture variants by swapping recurrence (GRU [9], LSTM [23], or feedforward) and attention (cross-attention, self-attention, additive [1], or mean pooling) independently while keeping all other hyperparameters fixed. Each configuration is run with and without the auxiliary loss (5 seeds each, 60 runs total). A MAAC-style variant places the auxiliary head on the centralized critic instead of the actor.

The pathology appears whenever *any* recurrence (GRU or LSTM) is paired with *any* learned attention mechanism (cross-attention or additive) and a constant auxiliary loss on the actor. Mean pooling, which has no learned attention parameters, is immune ($d = -0.30$). The MAAC-style critic-auxiliary configuration reverses the effect ($d = -0.79$): the auxiliary gradient flows through the critic rather

Table 10: Sample-efficiency: Final50 mean $\pm$ std (5 seeds) at each budget.

Config	500K	2.5M	5M	10M
Full	115.3 ± 9.3	453.3 ± 2.6	458.4 ± 4.5	470.3 ± 4.1
No Attn	119.8 ± 8.8	453.1 ± 5.6	461.9 ± 6.5	468.4 ± 5.1
No Aux	122.2 ± 16.5	460.8 ± 5.6	465.2 ± 5.4	468.2 ± 4.0
Neither	132.6 ± 17.7	459.9 ± 3.4	467.3 ± 6.8	470.3 ± 2.5

Table 11: λ sensitivity on Overcooked AA (Full VABL, 5 seeds). Final50 mean $\pm$ std. λ=0.01 is the stable sweet spot; λ=0.05 exhibits the directional-interference pathology; λ=0.001 is non-pathological in mean but has elevated cross-seed variance, indicating that the aux signal becomes too weak to regularize before it becomes safe to remove entirely.

λ	Budget	Final50	std	Best	Best std
0.001	10M	468.00	6.37	484.40	1.20
0.01	10M	472.23	1.67	484.40	1.20
0.05	10M	463.21	8.55	483.80	1.47
0.01	50M	468.47	8.73	485.00	0.00
0.05	50M	470.95	7.46	485.00	0.00

than the actor's belief encoder, so it corrupts value estimates (which are recomputed each epoch) rather than the policy representation (which accumulates damage through recurrence).

H Hyperparameters

I Figure Accessibility

All figures in this paper are designed for accessibility across viewing conditions. One author has color vision deficiency, which motivated a systematic approach to figure design:

- **Color palette.** We use the IBM colorblind-safe palette (magenta, blue, purple, orange, gold, teal), which maintains perceptual distinctness under deuteranopia, protanopia, and tritanopia. The pathological configuration (Full VABL) uses magenta throughout all figures for consistent identification.
- **Redundant encoding.** Every data series is distinguished by at least two independent visual channels. Dot plots use distinct marker shapes (cross, square, diamond, circle). Bar charts use hatching patterns (diagonal, dots, backslash, crosshatch) in addition to color. Line plots use distinct line styles (solid, dashed, dash-dot, dotted) with large markers.
- **Grayscale readability.** All figures remain interpretable when printed in black-and-white. We verified this by converting each figure to grayscale and confirming that all data series remain distinguishable through shape, pattern, and line style alone.
- **Font consistency.** Figure text (axis labels, legends, annotations) uses the same serif font family as the paper body to maintain visual coherence. We avoid matplotlib's default sans-serif rendering.
- **No duplicate titles.** Figures use LaTeX captions only, with no redundant matplotlib titles baked into the image. This avoids visual clutter and ensures consistent typography.

We encourage the MARL community to adopt similar accessibility practices. Approximately 8% of males and 0.5% of females have some form of color vision deficiency, and many readers access papers in printed grayscale. Relying solely on color to distinguish data series excludes these readers.

Table 12: Per-seed Final50 for the λ sweep at 10M on Overcooked AA. Best is reached consistently at $\sim$484 across all λ; only Final50 differentiates, confirming that λ affects late-training stability rather than peak performance.

λ	Seed 0	Seed 1	Seed 2	Seed 3	Seed 4
0.001	475.3	461.0	471.3	459.7	472.7
0.01	475.1	472.0	470.0	472.7	471.4
0.05	469.1	456.5	450.6	465.6	474.2

Table 13: Architecture sweep on Overcooked AA (25K episodes, 5 seeds each). Cohen's d: positive means aux hurts. The pathology generalizes across recurrence types and attention mechanisms but is absent with mean pooling and reverses when aux targets the critic.

Recurrence	**Attention**	**Aux loc.**	**+aux Final50**	**−aux Final50**	d	**Pathology?**
LSTM	cross-attn	actor	436.5 ± 22.1	454.0 ± 21.2	$+0.80$	**yes**
GRU	additive	actor	440.4 ± 21.8	524.8 ± 116.5	$+1.01$	**yes**
GRU	self-attn	actor	527.6 ± 145.2	544.4 ± 156.1	$+0.11$	no‡
GRU	mean pool	actor	431.7 ± 25.3	422.8 ± 33.2	-0.30	no
FF (none)	cross-attn	actor	588.1 ± 195.3	609.6 ± 208.7	$+0.11$	no‡
GRU	cross-attn	**critic**	546.6 ± 156.6	458.1 ± 21.5	-0.79	**reversed**

‡Both conditions exhibit high intrinsic cross-seed variance ($\sigma > 145$) in both the +aux and −aux cells, making the aux effect undetectable.

NeurIPS Paper Checklist

1. **Claims**

   Answer: [Yes]

   Justification: The abstract and introduction state five contributions, each supported by specific experimental evidence (Section 6) and theoretical analysis (Section 4). Scope limitations are explicitly discussed: the pathology is environment-dependent (present on Overcooked, absent on MPE) and specific to co-learning non-stationarity (absent in supervised CIFAR-100).

2. **Limitations**

   Answer: [Yes]

   Justification: Section 8 discusses environment scope (Overcooked + MPE, not SMAC/Hanabi), theoretical simplifications (quadratic loss, i.i.d. noise assumptions in Proposition 1), the cross-domain negative result, and the fact that VABL is not state-of-the-art in absolute reward.

3. **Theory assumptions and proofs**

   Answer: [Yes]

   Justification: Proposition 1 states all assumptions (quadratic loss near convergence, i.i.d. noise, spectral radius condition). A proof sketch appears in Section 4; the full proof is in Appendix A. We explicitly note that the proposition is a simplified model whose qualitative predictions we verify empirically.

4. **Experimental result reproducibility**

   Answer: [Yes]

   Justification: All hyperparameters are in Appendix H. Per-seed values are in Appendix C. The codebase uses JAX with fixed seeds; all runs are deterministic given the same seed and hardware. We plan to release code and data upon acceptance.

5. **Open access to data and code**

   Answer: [Yes]

   Justification: We will release the full codebase (JAX implementations of all 7 methods, training scripts, analysis scripts), all raw result JSONs (>200 runs), and a one-command reproduction script. An anonymized repository will be provided during review.

Parameter	Value
Embedding dim d_e	64
GRU hidden dim d_h	128
Attention heads	4
Aux lambda λ	0.05 (constant, unless annealed)
PPO [46] clip ϵ	0.2
PPO epochs	10
GAE λ [45]	0.95
Discount γ	0.99
Learning rate	5×10^{-4} (Adam [28], ϵ=10^{-5})
Gradient clip	10.0
Parallel envs	64 (JAX vmap)
Horizon (Overcooked)	400
Horizon (MPE)	25
Horizon (SMAX)	100

Table 14: Shared hyperparameters across all VABL configurations and baselines.

6. **Experimental setting/details**

   Answer: [Yes]

   Justification: Section 5 specifies environments, training budgets, number of seeds, vectorization setup, and evaluation metrics. Appendix H provides all hyperparameters. Appendix E provides multi-budget curves.

7. **Experiment statistical significance**

   Answer: [Yes]

   Justification: All MARL experiments use 5 random seeds. We report population standard deviation throughout and Cohen's d for effect sizes. Per-seed values are in Appendix C. The vision cross-domain test uses 1 seed (noted as a limitation).

8. **Experiments compute resources**

   Answer: [Yes]

   Justification: All experiments run on a single NVIDIA RTX 5090 (32GB). Per-run wall times are $\sim$8.5 minutes for 10M-step Overcooked runs and $\sim$27 minutes for 100K-episode MPE runs, with 64 vectorized environments via JAX vmap. Total compute across all experiments: $\sim$200 runs $\times$ $\sim$10 minutes $\approx$ 33 GPU-hours. The vision experiment uses a separate RTX 3060 Ti for smoke testing.

9. **Code of ethics**

   Answer: [Yes]

   Justification: This work studies optimization dynamics in cooperative multi-agent systems using standard benchmarks (Overcooked, MPE, CIFAR-100). No human subjects, private data, or dual-use concerns apply.

10. **Broader impacts**

    Answer: [N/A]

    Justification: This is foundational research on optimization pathologies in multi-agent reinforcement learning. The findings are methodological (how to design loss functions) and do not have direct negative societal applications. The practical recommendation (anneal or stop-gradient auxiliary losses) improves training reliability without introducing new risks.

11. **Safeguards**

    Answer: [N/A]

    Justification: The paper does not release pre-trained models, scraped datasets, or generative systems. The released code trains small cooperative agents on game benchmarks with no misuse potential.

12. **Licenses for existing assets**

Answer: [Yes]

Justification: We use Overcooked-AI (MIT license, Carroll et al. 2019), JaxMARL (Apache 2.0), CIFAR-100 (research use), and standard JAX/Flax/PyTorch libraries. All are properly cited.

13. **New assets**

    Answer: [Yes]

    Justification: We release a benchmark dataset of >200 training runs with raw episode rewards, and JAX implementations of 7 cooperative MARL methods. Documentation and reproduction instructions will accompany the release.

14. **Crowdsourcing and research with human subjects**

    Answer: [N/A]

    Justification: No human subjects or crowdsourcing involved.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Answer: [N/A]

    Justification: No human subjects involved.

16. **Declaration of LLM usage**

    Answer: [N/A]

    Justification: LLMs were not used as a component of the core methodology. LLMs were used for writing assistance and code generation during the research process, which does not require declaration per NeurIPS policy.